

Introduction to Information Technology: The Future Now

Chapter 1





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Chapter 1 Topics

UNIT 1A: The Mobile World, Information Technology, & Your Life

1.1 Information Technology & Your Life: The Future Now

1.2 Information Technology Is Pervasive: Cellphones, Email, the Internet, & the E-World

UNIT 1B: The Basics of Information Technology

1.3 The Practical User: How Becoming tech Smart Benefits You

1.4 The “All-Purpose Machine”: The Varieties of Computers

1.5 Understanding Your Computer

1.6 Where Is Information Technology Headed?





UNIT 1A: The Mobile World, Information, & Your Life

- As the result of developments in information technology, *smartphones* and *tablet computers* are changing nearly everything we do.
- *Information technology* refers to any technology that helps produce, manipulate, store, communicate, and/or disseminate information.



1.1 Information Technology & Your Life

The Future Now



Two Parts of IT: Computers & Communications

Information technology affects almost all aspects of our lives, including education, health, finance, recreation and entertainment, government, jobs and careers, and your personal life.

- Part 1: Computer Technology

A **computer** is a programmable, multiuse machine that accepts data—raw facts and figures—and processes, or manipulates, it into information we can use.



- Part 2: Communications Technology

Communications technology, also called *telecommunications technology*, consists of electromagnetic devices and systems for communicating over any distance.



Education: The Promise of More Interactive & Individualized Learning

Education has become heavily involved in information technology.

- Information technology can be used for:
 - Personalizing students' education
 - Automating tedious and rote tasks and managing classes
 - Reducing instructors' workload (course-management software)
 - Graphical presentations (e.g., PowerPoint)
- What is misuse?
 - Text messaging or emailing friends during class
 - Surfing the Internet for entertainment
 - Doing assignments for other classes
 - Sharing answers



Education: The Promise of More Interactive & Individualized Learning



- *Online Learning*, or *distance learning*, is becoming common.
 - Not all online schools/courses are accredited; students should check.
 - Online courses are less expensive than traditional courses.
 - Distance learning is available to students in rural areas.
- *Tutoring, simulation*, and *avatars* are also aspects of IT in education:



Health: High-Tech for Wellness

Computers are playing important roles in our personal lives.

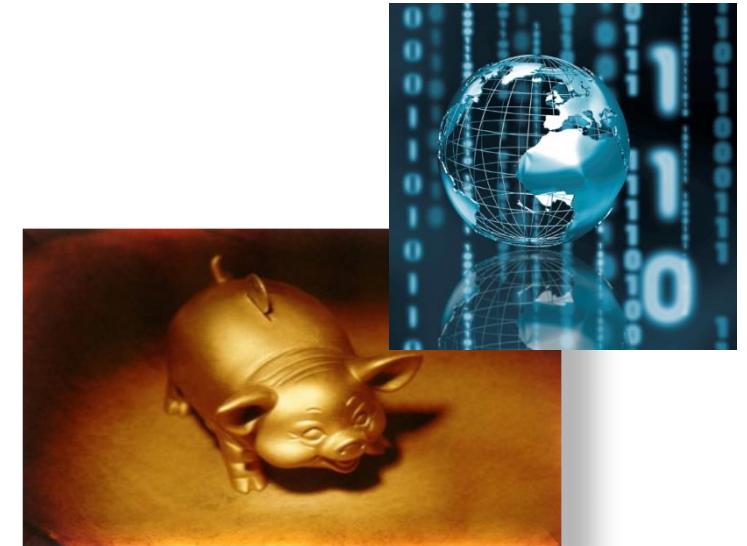


- **Telemedicine:** Medical care via telecommunications lets doctors treat patients from far away.
- 3D Computer models allow accurate tumor location inside a skull; X rays, MRIs, CT scans can be done remotely.
- **Robots**—*automatic devices that perform functions ordinarily performed by human beings*—permit precise microsurgery.
- Health websites provide medical information.
- Many health records are stored electronically.

Money & Business: Toward the Cashless Society?

Information technology is reducing the use of traditional money.

- **Virtual** means something that is created, simulated, or carried on by means of a computer or a computer network.
- Virtual money includes cash-value cards, automatic transfers, and digital money
 - “Electronic wallets” (e.g., PayPal)
 - Electronic payroll deposit
 - Online bill paying via debit and credit cards
 - Micropayments for online products and to help charities



Money & Business: Toward the Cashless Society?

- Smartphones are used for “showrooming” and shopping.
- Technology can also be used to telecommute and to start businesses and earn money.



Government & Electronic Democracy: Participating in the Civic Realm

Information technology is helping governments to deliver services and is affecting political activism.

- IT can help governments to improve services, including police services, which use **databases**, computer systems with a collection of interrelated files.
- Online voting is becoming common.
- Information is easier to disseminate.
- Watchdog websites are growing.
- Easier fund raising from small donors.

BUT:

- Gerrymandering is becoming easier—redrawing voting districts for partisan advantage.
- Voting machine problems can occur.
- Invasion of privacy is becoming an important issue.



Jobs & Careers

People now use computers to post résumés and find jobs.

- IT is used in starting new business ventures.
- IT is used to prepare résumés and find jobs on many websites.
- To help find jobs, participate in social media and write comments on **blogs** (weblogs), frequently updated sites on the web intended for public consumption that contain a writer's observations, opinions, images, and links to other websites.

(But be aware of privacy issues and DON'T POST inappropriate pictures or text!) (Internet postings live forever!)





Jobs & Careers

- Basic computer skills are needed for most jobs:
 - Know how to use a keyboard.
 - Use email.
 - Be able to use a word processor (usually Microsoft Word).
 - Know basic spreadsheet and database software skills.
 - Understand the basics of file sizes, computer memory limitations, and network arrangements.
 - Know what the basic computer system components are.



Discussion Question: Can anyone think of a career that does not require computer skills at all?



Your Personal Life

Computers are playing important roles in our personal lives.

- **Online relationship sites**, or *online dating sites*, provide electronic forums that people may join in the hope of meeting compatible companions or mates.
- “Digital is embedded into the fiber of every aspect of our culture and our personal lives” – public safety and security; in the home; entertainment; finance; communications; traveling; shopping; medical care; and so on.



Discussion Question: Can anyone think of an area of life NOT affected by IT?



1.2 Information Technology Is Pervasive

Cellphones, Email, the Internet, & the E-World

The Phone Grows Up

The telephone is not what it used to be.

- 1973: First cellphone call
- Mobile phone use estimated to rise to 7 billion users in 2014
- Today's smartphones can:
 - Make voice calls
 - Connect to Internet and web for all sorts of activities
 - Send and receive text messages
 - Take and send pictures and download music and video
 - Obtain news and TV programs
 - Scan special barcodes that take users to a website
 - Provide maps
 - Do research
 - Pay for products and services

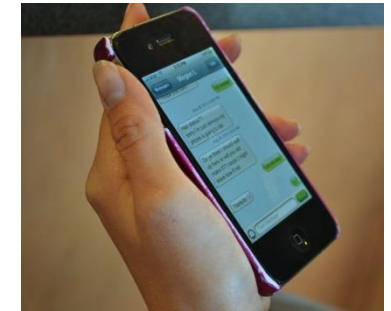


Email's Mass Impact



Email revolutionized communication, and has many benefits, but in many areas it is being supplanted by texting.

- Introduced in 1981
- Reached 10 million users in about one year
- 1998 surpassed hand-delivered mail
- In 2013 about 144 billion messages per day
- In business, at least, email requires writing skills
- For personal activities, texting is replacing email
 - **Texting**, or *text messaging*, is sending and receiving short written messages between mobile phones or other portable or fixed devices



The Internet, the World Wide Web, & the “Plumbing” of Cyberspace

The net, the web, and cyberspace are not the same things.

- **Cyberspace**
 - Term coined by William Gibson in *Neuromancer* (1984) to describe a futuristic computer network people “plugged” into directly with their brains
- Now the term **cyberspace** encompasses not only the online world and the Internet in particular but also the whole wired and wireless world of communications in general.



The Net & Web Defined



- The **Internet** (the “Net” or “net”) is a worldwide computer network that connects hundreds of thousands of smaller networks. These networks link educational, commercial, nonprofit, and military entities, as well as individuals.
- The **World Wide Web** (the “Web” or the “web” is an interconnected system of Internet computers (called *servers*) that support specially formatted documents in multimedia form. (The word *multimedia*, from “multiple media,” refers to technology that presents information in more than one medium, such as text, still images, moving images, and sound.)





UNIT 1B: The Basics of Information Technology

- As the result of developments in information technology, *smartphones* and *tablet computers* are changing nearly everything we do.
- *Information technology* refers to any technology that helps produce, manipulate, store, communicate, and/or disseminate information.

1.3 The Practical User

How Becoming tech Smart Benefits You



Being informed about information technology has practical payoffs.

- **Know** how to make better buying decisions
- **Know** how to fix ordinary computer problems
- **Know** how to upgrade equipment and integrate it with new products
- **Know** how to use the Internet effectively
- **Know** how to guard against online dangers
- **Know** how computer knowledge can advance your career



1.4 The “All-Purpose Machine”

The Varieties of Computers

All Computers, Great & Small: The Categories of Machines

Computers come in different sizes; they also function as clients and/or servers.

- There are five basic computer sizes.*

1. Supercomputers
2. Mainframes
3. Workstations
4. Microcomputers
5. Microcontrollers



1. Supercomputers

Supercomputers are used in very special situations.

- Priced from \$1 million to over \$350 million.
- High-capacity machines with thousands of processors that can perform more than several quadrillion calculations per second.
- Faster and largest computer available.
- Used for government census, weather forecasting, designing aircraft, scientific projects, etc.
- The Titan (U.S.A.) computer is currently the largest supercomputer.
- The next supercomputer generation may use **nanotechnology**.



2. Mainframes

Mainframe computers are used in many large businesses.

- Priced from \$5,000 to \$5 million
- Process billions of instructions per second
- Size is dependent on the use
- Water-cooled or air-cooled
- Used to be called *midsize computers*
- Used by banks, airlines, colleges, and the like for millions of transactions



3. Workstations

Workstations are used for graphics, special effects, and certain professional applications.

- Expensive, powerful personal computers
- Used for scientific, mathematical, engineering, computer-aided design (CAD), computer-aided manufacturing (CAM) applications
- Used for designing cars, drugs, movie special effects
- Are usually connected to a network



4. Microcomputers

Microcomputers are used by individuals as well as businesses, and they can be connected to networks of larger computers. There are many types of microcomputers.

- Personal computers that cost \$500 to over \$5,000
- Used either stand-alone or in a network
- Types include: desktop, tower, notebooks (laptops), netbooks, tablets, mobile devices, personal digital assistants (PDAs), and e-readers



Types of Microcomputers (1)

- Desktop and tower PCs



Desktop PC



Tower PC

- Notebooks & netbooks



- Tablets



Types of Microcomputers (2)

- Mobile devices & PDAs



- E-readers



- Also called *embedded computers*, **microcontrollers** are tiny, specialized microprocessors inside appliances and automobiles. They are in microwaves, programmable ovens, blood-pressure monitors, air bag sensors, vibration sensors, MP3 players, digital cameras, keyboards, car systems, etc.

Servers



- The word *server* describes the way a computer—whether mainframe, workstation, or PC—is used.
- A **server**, or *network server*, is a central computer that holds collections of data (databases) and programs for connecting or supplying services to PCs, workstations, and other devices, which are called **clients**. These clients are linked by a wired or wireless network. The entire network is called a *client-server network*.
- Purpose: Hold data and programs for clients to access and to supply services for clients.



1.5 Understanding Your Computer

How Computers Work: Three Key Concepts

All computer users must understand three basic principles: (1) Data is turned into information; (2) hardware and software have their own specific functions; and (3) all computers involve input, processing, storage, and output, plus communications.

1. Purpose of a computer: Turning data into information

- **Data:** the raw facts and figures
- **Information:** data that has been summarized or otherwise transformed for use in decision making

2. Hardware vs. software

- **Hardware** = the machinery and equipment in a computer system
- **Software** (programs) = the electronic instructions that tell the computer how to perform a task

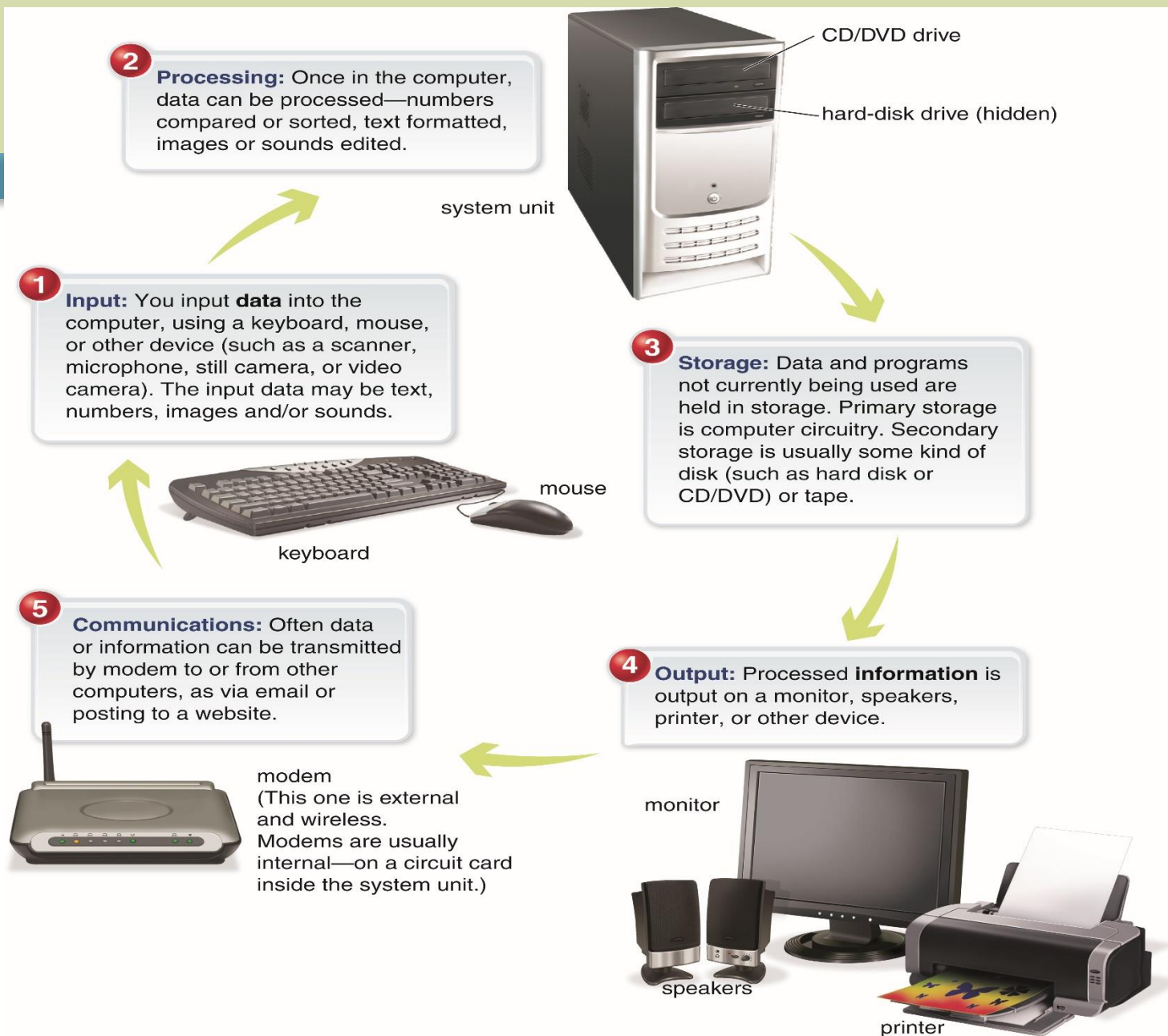
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3. The basic operations of a computer: All computers use 4 basic operations + communications:

- **Input:** What goes into the computer system
- **Processing:** The manipulation a computer does to transform data into information
- **Storage:**
 - **Primary storage**, or **memory**, is temporary storage for data waiting to be processed
 - **Secondary storage** is permanent storage: media such as hard disk, DVDs, and CDs
- **Output:** What comes out—the results of processing, such as on the screen, printouts, sound
- **Communications:** Sending and receiving data



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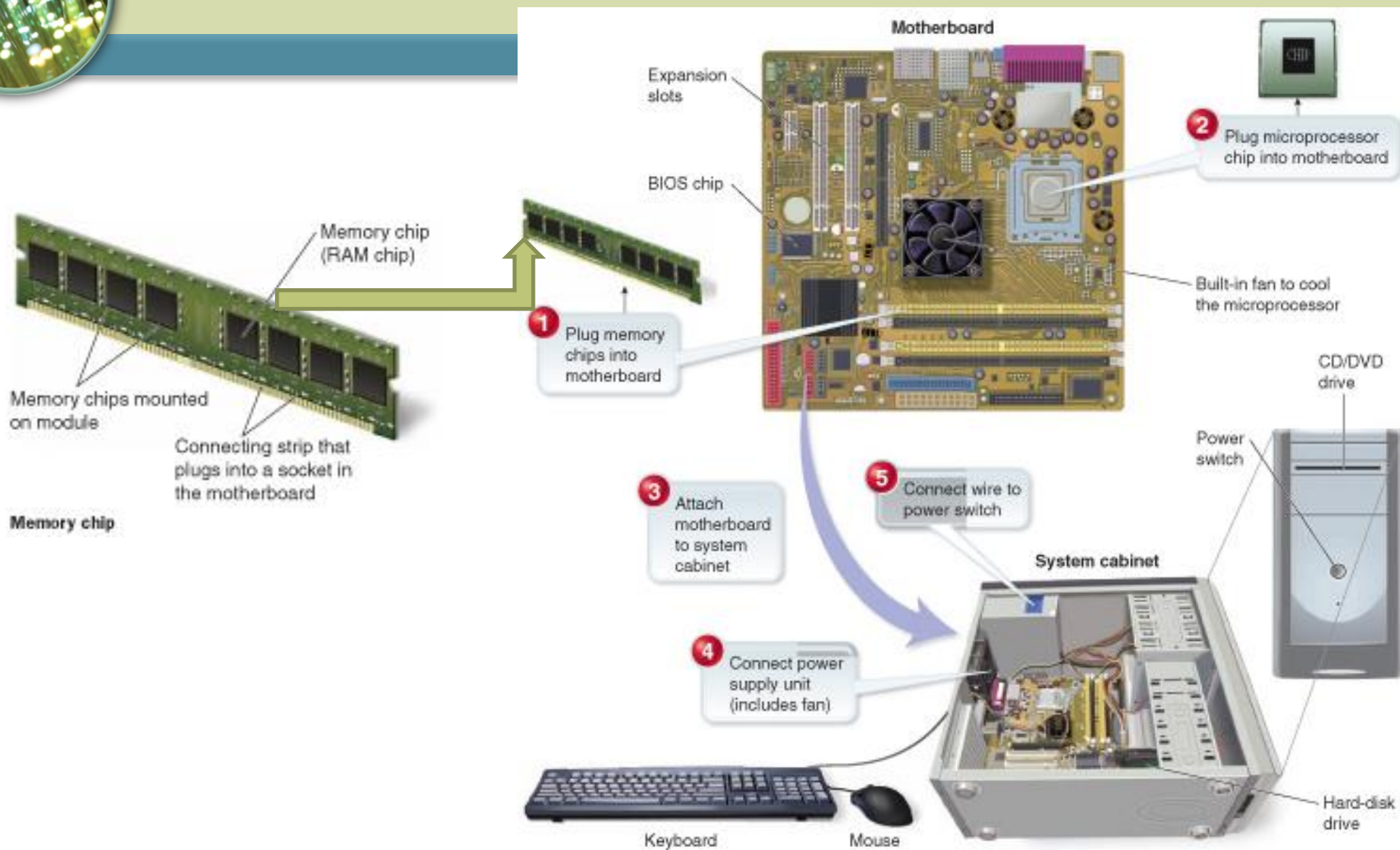
Customizing a Desktop Computer

- What would you need?
 - Keyboard & mouse (**input hardware**)
 - Inside the system cabinet (**processing & memory hardware**)



- Case and power supply
- Processor chip – the central processing unit (CPU)
- Memory chips – random access memory (RAM) or primary storage
- Motherboard – the system board, the main circuit board, with expansion slots to plug in components



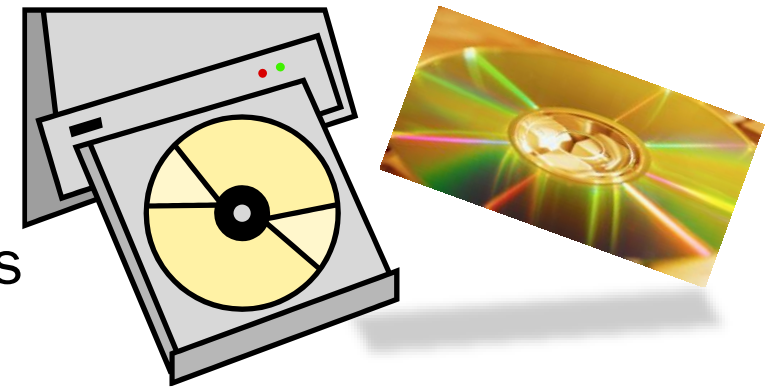


Putting the
components
together
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- **Storage Hardware:** Hard Drive, CD/DVD Drive

- Storage capacity is represented in bytes
 - 1 byte = 1 character of data
 - 1 kilobyte = 1,024 characters
 - 1 megabyte = 1,048,576 characters
 - 1 gigabyte = over 1 billion characters
 - 1 terabyte = over 1 trillion characters
 - 1 petabyte = about 1 quadrillion characters





- **Output hardware**



- Video
- Sound cards
- Speakers
- Monitor
- Printer

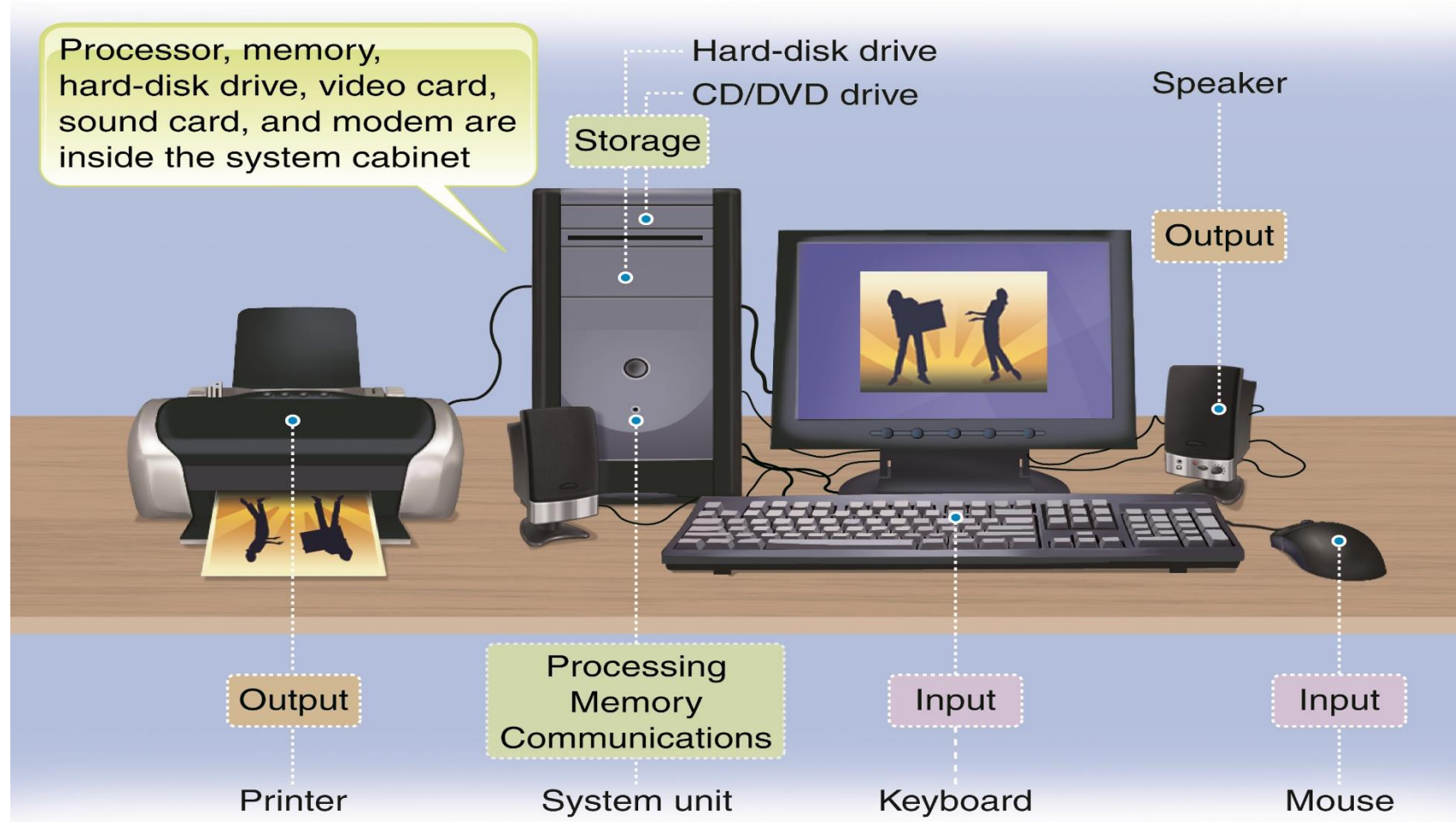


- **Communications hardware**

- Modem



Basic PC System

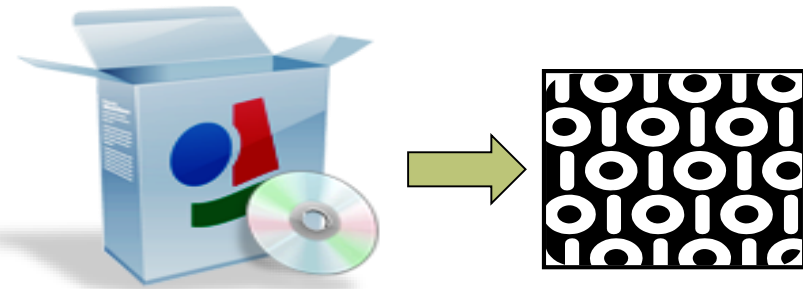


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Software

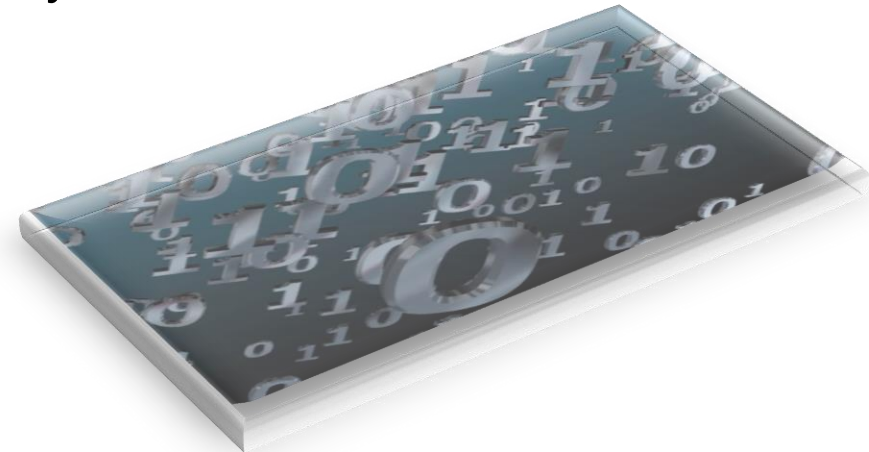
Computers use two basic types of software: system software and application software.

- **System Software—enables** the computer to perform essential operating tasks and makes it possible for application software to run.
 - Most important part: operating system (OS)
 - Some operating system options
 - Windows
 - Unix
 - Linux
 - Mac OS



Software

- **Application Software**—enables you to perform specific tasks—solve problems, perform work, or entertain yourself.
 - Compatibility: Application software is specific to the system software you use.
 - Linux applications won't work on Windows.
 - Windows applications won't necessarily work on Linux.





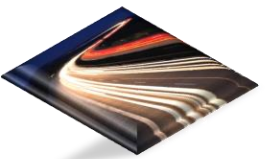
1.6 Where Is Information Technology Headed?



Computers are headed in three basic directions—



miniaturization,



faster speeds, *and*



greater affordability

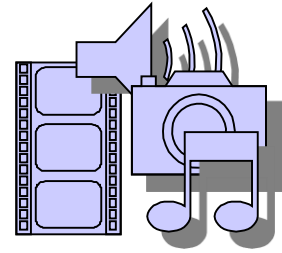
—and communications are improving



connectivity,



interactivity, *and support of*



multimedia.

When Computers & Communications Converge: Five Results

Convergence—combination of several industries

- Computers
- Communications
- Consumer electronics
- Entertainment
- Mass media

1. Portability
2. Personalization
3. Collaboration
4. Cloud computing
5. Artificial intelligence





- **Cloud computing** basically means that instead of storing your software and/or data on your own PC or your own company's computers, you store it on servers on the Internet.
- **Artificial intelligence (AI)** refers to a group of related technologies used for developing machines to emulate human qualities, such as learning, reasoning, communicating, seeing, and hearing. Much of AI is based on the use of **algorithms**, formulas or sets of steps for solving particular problems. AI deals with **Big Data**, data that is so large and complex that it cannot be processed using conventional methods,





Ethics

Many important ethical issues are involved in the use of information technology.

- **Ethics** is the set of moral values or principles that govern the conduct of an individual or a group.
- 3 ethical considerations resulting from development of IT:
 - Speed & scale affect security and personal privacy
 - Unpredictability—IT can be less predictable and reliable than other technologies
 - Complexity—computer systems can be unmanageable